Aegis OS: A Transactional Multi-Tenant Operating System Runtime for Eliminating Authorization Fatigue in Autonomous Coding Agents

Hui Zhao
Heurion Research
huizhao@heurion.org

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Abstract

Autonomous coding agents such as Google Antigravity (`agy`), Anthropic Claude Code (`claude`), and OpenAI Codex CLI (`codex`) execute multi-step software engineering workflows on developer workstations and cloud hosts. Existing agent execution environments face a *Security–Autonomy Dilemma*: operators must choose between per-action confirmation prompts, which interrupt workflows on 45%–57% of tool calls in our benchmark trajectories and induce severe authorization fatigue, or unconfined execution (`-yolo` / `-dangerously-skip-permissions`), which exposes host files, SSH/cloud credentials, and API tokens to prompt injection and hallucination risks. Furthermore, concurrent agents editing a shared working tree clobber one another's changes, while passing large artifacts inline through LLM context windows incurs high token overhead.

We present **Aegis OS** (`aegis-os`), a transactional, multi-tenant Agent Operating System Runtime written in Rust and exposed via the Model Context Protocol (MCP) and the Google Agent2Agent (A2A) protocol. Aegis OS shifts the authorization boundary from individual tool calls to *transactional workspace boundaries*. Each agent operates inside a kernel-enforced OS sandbox (evaluated on macOS Seatbelt `sandbox-exec`, with a Linux Bubblewrap namespace backend and microVM control-plane interface) backed by an APFS Copy-on-Write (CoW) shadow workspace (`libc::clonefile`), an identity-pinned HTTP Egress Proxy with boundary secret injection, and an A2A-compliant Inter-Agent Communication (IPC) bus paired with Content-Addressable Storage (CAS). Inside its isolated CoW workspace and organization IPC bus, an agent executes with zero confirmation prompts; human approval or TTL-scoped session leases are required only when committing diffs to the host via an LCS-based Diff3 3-Way Merge Engine. Across 1,500 runtime tool calls spanning five trajectory templates and 40 end-to-end adversarial attacks, Aegis OS reduces human confirmation prompts by **94.19%–99.88%**, lowers the attack success rate on evaluated vectors from **100.0% to 0.0%**, clones 1,000-file (78.1 MB) workspaces in **6.28 ms** (**18.0× faster** than full copy), and preserves 100% of disjoint multi-agent edits while catching 10/10 overlapping line conflicts.

1 Introduction

The paradigm of AI-assisted software engineering has shifted from passive single-line code completion to autonomous, long-horizon coding agents—such as Google Antigravity (`agy`), Anthropic Claude Code (`claude`), OpenAI Codex CLI (`codex`), SWE-agent [26], and OpenHands [21]. Unlike conversational assistants, coding agents iteratively explore repositories, modify files across multiple modules, invoke compilers and test suites, query external APIs, and coordinate subtasks with peer agents [13, 3].

Despite advances in model reasoning, the execution substrate for local coding agents remains caught between usability and safety. When agent CLIs mutate the developer's live working tree directly, every file write (`fs_write`, `fs_edit`) and shell command (`proc_exec`) carries immediate side effects on the host. Consequently, agent frameworks rely heavily on *action-level human-in-the-loop (HITL) confirmation prompts*.

1.1 The Security–Autonomy Dilemma

In practice, action-level prompting and existing in-place CLI sandboxes exhibit three systemic limitations:

1. **Severe Authorization Fatigue:** A 30-step bugfix, refactoring, or security-audit trajectory typically invokes 13 to 18 mutating or state-changing operations. Prompting the developer on 45%–57% of tool invocations (as measured in §5) fragments autonomous workflow continuity to streaks of 4–7 steps and habituates users to reflexively

approve prompts without inspecting arguments—a well-documented failure mode in usable security [9, 2].

2. **Risk of Unconfined "YOLO" Mode and In-Place Mutation:** To avoid constant interruptions, developers frequently enable bypass flags (`-dangerously-skip-permissions`, `-yolo`) or rely on lightweight in-place CLI sandboxes (e.g., Claude Code or Codex native Seatbelt/Landlock modes). In unconfined mode, an indirect prompt injection [11, 8, 27] can overwrite files outside the repository, read host credentials (`~/.ssh/id_rsa`, `~/.aws/credentials`, `~/.kube/config`), or exfiltrate raw API keys (`$GITHUB_TOKEN`) from process environment variables. Even when native CLI sandboxes restrict writes to the project directory, agents still mutate the live host working tree in place without transactional rollback or boundary secret injection.

3. **Multi-Agent Workspace Collisions and Context Bloat:** Developers increasingly run heterogeneous agents in parallel (e.g., `agy` for architecture, `claude` for backend logic, and `codex` for test generation) [24, 12]. When sharing a live working directory without manual `git worktree` management, concurrent edits overwrite one another's changes, while passing large specifications inline through JSON/-text messages bloats LLM context windows.

1.2 Key Insight: Transaction-Boundary Security

Inspired by operating system transactions [18, 17] and layered copy-on-write filesystem sandboxes [15, 23], our key insight is that *authorization prompts should not guard speculative intermediate actions inside an isolated state space; they should guard irreversible boundary crossings into the host repository or untrusted external networks.*

In Aegis OS, each agent task executes as a three-step workspace transaction (Table 1):

1. **Step 1 — Instant CoW Shadow Fork (0.7–6.3 ms):** When an agent session starts, Aegis OS clones the host repository into an isolated Copy-on-Write (CoW) shadow workspace via APFS `libc::clonefile`, sharing unchanged disk blocks with zero data copying.

2. **Step 2 — Green-Zone Speculation (0 Prompts):** Inside its shadow workspace, the agent freely edits files, runs compilers and tests inside a kernel OS sandbox, coordinates with peer agents over A2A/IPC, and queries allowlisted APIs using opaque secret placeholders (`$AOS_SECRET_*`). Because the live host tree cannot be modified and plaintext credentials never enter the sandbox, Aegis OS auto-allows 100% of in-sandbox operations with **zero confirmation prompts**.

3. **Step 3 — Boundary Commit Gate (1 Prompt or Lease):** Only when the agent finishes testing and invokes `workspace_commit` does Aegis OS validate a two-phase 3-way merge against the live host repository and request human approval (or consume a time-bounded session lease).

1.3 Contributions

We design, implement, and evaluate **Aegis OS** (`aegis-os`), a multi-tenant Agent Operating System Runtime in Rust:

- **Transactional CoW VFS with Two-Phase LCS Diff3 Merge (§3.1):** `aos-vfs` uses APFS `libc::clonefile` syscalls to provision isolated shadow workspaces in metadata time (0.71–6.28 ms for 100–1,000 files), supports millisecond snapshots and rollbacks while preserving compiler build caches (`CARGO_TARGET_DIR`) outside the rollback tree, and implements a two-phase Longest Common Subsequence (LCS) Diff3 3-Way Merge Engine [14] that reconciles concurrent edits from heterogeneous agents while aborting cleanly on overlapping line conflicts.

- **Kernel Sandbox and Identity-Pinned Egress Proxy (§3.2):** `aos-sandbox` and `aos-egress` enforce mandatory access control via macOS Seatbelt (`/usr/bin/sandbox-exec` with file-write, credential-read, AppleEvent, and exact proxy-port network confinement) and a Linux Bubblewrap (`bwrap`) namespace driver, paired with a TCP-pinned HTTP Egress Proxy that injects `$AOS_SECRET_*` tokens at the network boundary.

- **A2A-Compliant Inter-Agent IPC Bus and Reference-Based CAS (§3.3):** `aos-ipc` aligns inter-agent coordination with the Google Agent2Agent (A2A) protocol [10], providing global peer discovery with kernel PID liveness pruning, dynamic `AgentCard` synthesis (`/.well-known/agent-card.json`), stateful A2A `Task` lifecycle tracking (`POST /a2a`), P2P mailboxes, topic Pub/Sub, and Content-Addressable Storage (`artifact://<org>/sha256:<hash>`) that reduces in-transit IPC payload tokens by 99.85%.

- **Cedar-Inspired Policy Engine with Commit-Time Poisoning Guards (§3.4):** `aos-policy`

Table 1: **Comparison of Agent Execution Substrates** across isolation mechanisms, authorization friction, workspace storage, and multi-agent coordination primitives.

Execution Substrate	Workspace Setup	N-Agent Disk Cost	Prompt Rate (IPR)	OS Kernel MAC	Boundary Secret Inject	Transactional 3-Way Merge	Native IPC + CAS
Host-Interactive (Unconfined CLI)	In-Place (0 ms)	Shared Tree	High (45%–57%)	× (Prompt Only)	× (Raw Env Vars)	× (In-Place Write)	×
Host-YOLO (`-skip-permissions`)	In-Place (0 ms)	Shared Tree	0.0%	× (Unconfined)	× (Raw Env Vars)	× (Silent Clobber)	×
Native CLI Sandbox (`claude/codex`)	In-Place (∼6 ms)	Shared Tree	Low–Med	✓ (Seatbelt/Landlock)	× (Env in Sandbox)	× (Shared Working Tree)	×
Container-per-Agent (Docker) [13, 21]	Container Init	Per-Container Layer	Low	✓ (Linux NS/cgroups)	× (Env in Container)	× (Isolated Volume)	×
AIOS (LLM Scheduler Kernel) [16]	In-Place	Shared Tree	Medium	× (No OS MAC)	× (No Egress Proxy)	× (No CoW Merge)	Partial
Aegis OS (Ours)	**0.71–6.28 ms (CoW)**	**Modified Files Only**	**0.07%–3.33%**	✓ (`sandbox-exec/bwrap`)	✓ (`$AOS_SECRET_*`)	✓ (2-Phase LCS Diff3)	✓ (`aos-ipc`)

implements a five-stage authorization hierarchy inspired by Cedar [6], combining host red-line protection and commit-time sensitive path guards (`.git/hooks`, `.envrc`, `.github/workflows`) with one-shot and TTL-scoped session leases.

- **Empirical Evaluation (§5):** We evaluate Aegis OS on Apple Silicon across 1,500 runtime tool calls, 40 end-to-end process/socket/gateway red-team attacks, multi-scale APFS CoW working-set benchmarks, and multi-agent Rust compilations (`cargo test` and `cargo check`).

2 Threat Model and Evaluation Metrics

2.1 System and Adversary Model

We consider a multi-agent deployment where autonomous coding agents operate within one or more tenant organizations. Each agent session is identified by its organization tenant (e.g., `aegis-os`), unique agent principal ID (e.g., `agy-agent`, `claude-code`, `codex-cli`), and role.

Trusted Computing Base and Enforcement Assumption. We trust the host operating system kernel (macOS XNU/Seatbelt or Linux kernel namespaces) and the Aegis OS Runtime binary (`aegis-os`), which executes outside the agent sandbox. Because MCP is an additive tool protocol, our security guarantees assume that the host agent client is configured to route filesystem mutations and shell execution through `aegis-os` (i.e., built-in unconfined host tools such as raw `Bash` or host `Write` are either disabled in client permissions or intercepted by client `PreToolUse` hooks, as detailed in §4).

Adversarial Threat Vectors (T1–T5). We assume the LLM driving any agent is untrusted and may issue malicious or erroneous tool calls due to:

- **T1 (Indirect Prompt Injection):** Malicious instructions embedded in repository files, commit logs, issue trackers, or web pages read by the agent [11, 8].
- **T2 (Host Path Escape and Tampering):** Attempting to write files outside the workspace (e.g., `../../etc/hosts`, `~/.zshrc`, or sibling repositories) or staging commit-time hooks (`.git/hooks/pre-commit`, `.envrc`).
- **T3 (Credential Harvesting):** Attempting to read sensitive host credentials (`~/.ssh/id_rsa`, `~/.aws/credentials`, `~/.gnupg`, `~/.kube/config`, `~/.npmrc`, or macOS Keychains).
- **T4 (Network Secret Exfiltration):** Attempting to transmit API tokens or source code to unauthorized external domains (`evil-exfil.org`, `webhook.site`) via HTTP requests or direct socket bypass (`curl -noproxy '*'`).
- **T5 (Cross-Tenant Privilege Escalation):** Attempting to read CAS artifacts or inject IPC messages into a different tenant organization.

Non-Adversarial Concurrency and Reliability Hazards (H1–H2). Even in benign settings, multi-agent workflows face: **(H1) Concurrent Lost Updates**, where parallel agents overwrite one another's edits on shared files; and **(H2) Multi-File Hallucinated Corruption**, where an agent breaks syntax across dozens of modules during speculative refactoring.

2.2 Evaluation Metrics

We evaluate authorization friction and security using four direct operational metrics:

- **Interruption Prompt Rate (IPR):** The percentage of runtime tool calls that halt execution to request interactive human confirmation (Human Prompts/Total Tool Calls × 100%).
- **Modeled Human Verification Wait Time (T_{human}):** The cumulative human review overhead across a workload, assuming a conservative average context-switch and inspection time of 5.0 s per confirmation prompt.
- **Maximum Autonomous Streak (MAS):** The longest contiguous sequence of tool calls executed without a human confirmation prompt.
- **Attack Success Rate (ASR):** The percentage of adversarial attacks that successfully modify host

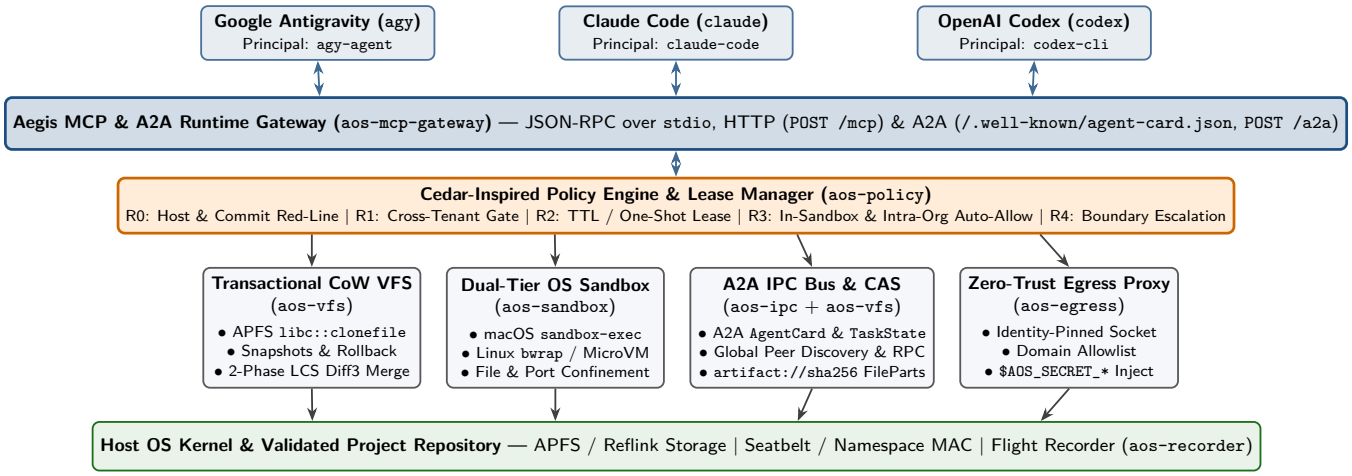


Figure 1: Architecture of the Aegis OS Runtime (`aegis-os`). Coding agents connect via MCP (`stdio` or HTTP daemon) as distinct principals within a tenant organization. The policy engine auto-allows in-sandbox CoW mutations, confined process executions, and intra-organization IPC/CAS operations with zero prompts, while guarding host commits and network egress.

files outside the workspace, read restricted credentials, exfiltrate secrets, or breach tenant isolation.

3 Aegis OS Runtime Architecture

Figure 1 illustrates the architecture of Aegis OS, and Table 2 summarizes its 35 MCP tools, 7 dynamic URI resources, and native Google A2A HTTP endpoints. Aegis OS follows an *"Environment Runtime Only"* design: it does not embed LLM planners or prompt templates, instead providing virtual filesystem transactions, OS process isolation, network boundary proxying, A2A-compliant inter-agent coordination, and policy enforcement.

3.1 Transactional CoW Virtual Filesystem and Two-Phase LCS Diff3 Merge (`aos-vfs`)

To eliminate confirmation prompts on file mutations without risking host corruption, `aos-vfs` isolates every agent principal inside an independent Copy-on-Write shadow workspace rooted at:

`<runtime_dir>/orgs/<org>/agents/<agent>/cow_workspace/upper_workspace/`

(where `<runtime_dir>` defaults to `/tmp/aegis-os-runtime` or `~/.aegis-os`), alongside a `baseline_snapshot/` directory recording the host repository state at session initialization or last commit, and a `snapshots/<snap_id>/` directory storing time-travel checkpoints.

APFS `clonefile` Shadow Provisioning. Copying a multi-megabyte repository per agent via user-space read/write loops scales linearly with total byte volume. On macOS (APFS), `aos-vfs` invokes the native XNU syscall `libc::clonefile(src, dst, 0)` on eligible directory subtrees and files (automatically skipping heavy build/VCS directories such as `.git`, `target`, and `node_modules`). Because `clonefile` duplicates only filesystem metadata extents and shares physical blocks until mutated, workspace initialization depends only on file count (0.71–6.28 ms for 100–1,000 files) and allocates physical storage only for modified files. Unlike `git worktree`, which requires a valid git commit history, shares the repository object store, and cannot snapshot dirty untracked trees in sub-millisecond time, `aos-vfs` operates transparently on any directory and supports instant nested checkpoints.

Time-Travel Snapshots, Rollback, and Build-Cache Preservation. Before speculative multi-file edits, an agent calls `workspace_snapshot(label)`, creating a CoW clone under `snapshots/<snap_id>/`. If compilation fails, `workspace_rollback(snap_id)` replaces `upper_workspace/` with a fresh CoW clone of the snapshot in milliseconds. To prevent rollbacks from invalidating incremental compiler caches, `aos-sandbox` redirects `CARGO_TARGET_DIR` to `agent_home/cargo_target` outside `upper_workspace/`.

Two-Phase LCS Diff3 3-Way Merge Engine. When multiple agents (e.g., `agy-agent`, `claude-code`, and `codex-cli`) concurrently edit the same file in their shadow workspaces and sequentially invoke `workspace_commit`, a naive file copy would overwrite earlier peer commits (Hazard H1). Aegis OS prevents

Table 2: **Aegis OS MCP Tool, Dynamic URI Resource, and Google A2A Interface** (35 MCP Tools and 7 Dynamic URI Resources implemented in `aos-mcp-gateway`).

Subsystem	MCP Tools / Dynamic URI Resources / A2A Endpoints	Default Policy Gate & Runtime Semantics
Transactional VFS (`aos-vfs`, 6 tools)	`fs_read`, `fs_write`, `fs_edit`, `fs_delete`, `fs_glob`, `workspace_status`	**Auto-Allow (0 Prompts)**: Confined to agent's CoW shadow tree
Time-Travel, Sync & Merge (`aos-vfs`, 6 tools)	`workspace_snapshot`, `workspace_diff`, `workspace_rollback`, `workspace_sync`, `workspace_commit`, `workspace_export_pr`	Snapshots/Rollback/Rebase-Sync **Auto-Allow**; `workspace_commit` & `workspace_export_pr` **Approval/Lease Gate**
OS Process Sandbox (`aos-sandbox`, 1 tool)	`proc_exec` (macOS Seatbelt `sandbox-exec` / Linux `bwrap` / MicroVM spec)	**Auto-Allow (0 Prompts)**: Writes restricted to CoW root; blocks credentials
Inter-Agent IPC & Claims (`aos-ipc`, 10 tools)	`ipc_send`, `ipc_recv`, `ipc_request`, `ipc_reply`, `ipc_ack`, `ipc_publish`, `ipc_subscribe`, `ipc_list_peers`, `coord_claim`, `coord_release`	**Auto-Allow Intra-Org & Workstation IPC**: Verified global peers (`<org>/<agent>`) allowed; file claims enforced
Google A2A Protocol (`aos-ipc`, 4 tools + HTTP)	`ipc_update_card`, `ipc_task_update`, `ipc_task_get`, `ipc_task_list` + `GET /.well-known/agent-card.json`, `POST /a2a`	**Auto-Allow Discovery & Task RPC**: `AgentCard` synthesis & stateful `A2ATaskState` lifecycle (`submitted`→`completed`)
Org CAS & KV (`aos-vfs`, 4 tools)	`org_artifact_publish`, `org_artifact_fetch`, `org_kv_put`, `org_kv_get`	**Auto-Allow Intra-Org**: Cross-org `artifact://` `FilePart` fetch requires policy boundary approval (**Ask**/**Deny**)
Zero-Trust Egress (`aos-egress`, 1 tool)	`egress_http_request` + Identity-Pinned TCP Proxy (`127.0.0.1:<port>`)	**Allowlisted Domains Auto-Allow**: Unlisted domains escalate; injects `$AOS_SECRET_*`
Policy & Leases (`aos-policy`, 3 tools)	`policy_list_escalations`, `policy_approve_escalation`, `policy_grant_lease`	Operator control plane for one-shot (`ttl=0`) or TTL-scoped `LeaseGrant` tokens
Dynamic MCP URIs (7 resources)	`aegis://workspace/status`, `aegis://ipc/inbox`, `aegis://org/peers`, `aegis://global/peers`, `aegis://a2a/agent-cards`, `aegis://a2a/tasks`, `aegis://org/shared/catalog`	Live state introspection for IDE status bars, global peer topology, and A2A task tables

lost updates via a two-phase Longest Common Subsequence (LCS) Diff3 3-Way Merge algorithm [14] comparing three versions of each modified file: the agent's recorded **Baseline** snapshot, the live **Host** file, and the agent's **Shadow** file:

- **Phase 1 (Validation & In-Memory Staging):** Aegis OS inspects all modified, added, and deleted files in the commit batch before touching any host file. If the live Host file is unchanged since the agent's Baseline, the Shadow file is staged for a fast-forward CoW clone. If another agent has already modified the Host file, Aegis OS computes line-level LCS diffs against the Baseline and splices non-overlapping edit hunks together automatically (for instance, if `agy-agent` inserts a 3-line helper at the top of `src/lib.rs` while `claude-code` modifies a function 40 lines below, the shifted line numbers are aligned and merged cleanly). If both agents modified the same baseline lines in incompatible ways, Phase 1 aborts the entire commit without modifying any host file and returns the exact conflicting line numbers.
- **Phase 2 (Atomic Commit Application):** Once all files in the batch pass Phase 1 validation, staged merges and CoW clones are written to the host repository and synced to `baseline_snapshot/`. Conversely, when a long-lived agent resumes a session after external host commits, `refresh_from_host()` rebases its shadow workspace onto the latest host state.

3.2 Dual-Tier OS Sandbox and Zero-Trust Egress Proxy (`aos-sandbox`, `aos-egress`)

While `aos-vfs` mediates MCP file tools, coding agents also spawn compilers, test runners, and shell scripts via `proc_exec`. To prevent child processes from bypassing workspace or network boundaries, `aos-sandbox` provides a pluggable `SandboxDriver` with two tiers:

1. **Tier 1 (`NativeOsSandboxDriver`, Workstation Kernel Enforcement):** On macOS, every `proc_exec` call compiles a canonicalized Seatbelt Mandatory Access Control profile (SBPL) and spawns the child process under `/usr/bin/sandbox-exec -p <profile>` with a sanitized environment (`env_clear()`). The generated profile blocks AppleEvent IPC (`(deny appleevent-send)`), denies all file writes globally (`(deny file-write*)`), allowlists writes strictly to the agent's CoW shadow workspace, private agent home, and organization publish directory, blocks reads to sensitive credential paths, and—when an egress proxy is bound—restricts outbound network sockets exclusively to the exact loopback port of the agent's proxy:

```
(version 1)
(allow default)
(deny appleevent-send)
(deny file-write*)
(allow file-write*
  (subpath "/private/.../upper_workspace")
  (subpath "/private/.../home")
  (subpath "/private/.../pub/agy-agent")
  (literal "/dev/null") (literal "/dev/zero") (literal "/dev
     /tty")
  (literal "/dev/random") (literal "/dev/urandom"))
(deny file-read*
  (regex #".*/\.ssh(/.*)?$")   (regex #".*/\.aws(/.*)?$")
  (regex #".*/\.gnupg(/.*)?$") (regex #".*/\.kube(/.*)?$")
  (regex #".*/\.npmrc$")       (regex #".*/\.pypirc$")
  (regex #".*/\.config/gh(/.*)?$")
  (regex #".*/Library/Keychains(/.*)?$")
  (literal "/etc/shadow"))
(deny network-outbound)
(allow network-outbound (remote ip "localhost:51691"))
```

Because `(deny network-outbound)` is enforced by the XNU kernel socket filter and allows only `localhost:<proxy_port>` (without permitting wildcard localhost ports or `mDNSResponder`), a child process running `curl -noproxy '*'` or opening raw sockets to external IPs or other local services fails immediately with `Operation not permitted`. On Linux, `NativeOsSandboxDriver` constructs Bubblewrap (`/usr/bin/bwrap`) arguments with `-unshare-all` (unsharing user, mount, PID, IPC, and network namespaces when an egress proxy is bound), read-only root binds (`-ro-bind / /`), writable binds only for the shadow workspace and agent home, and empty `-tmpfs` overlays masking `~/.ssh`, `~/.aws`, `~/.gnupg`, and `~/.kube`.

2. **Tier 2 (`MicroVmSandboxDriver`, Cloud Control-Plane Interface):** Provides the configuration generator (`virtio-fs` mount tags, vCPU/memory limits) and Scale-to-Zero `pause()`/`resume()` state machine for cloud microVM backends (`libkrun` / Firecracker [1]).

Identity-Pinned Egress Proxy with Boundary Secret Injection. To mitigate credential ex-

filtration (Threat T4), agents inside Aegis OS do not receive raw API keys in environment variables. Instead, they receive opaque placeholders such as `$AOS_SECRET_GH`. Outbound HTTP traffic is routed through a local asynchronous L7 HTTP proxy (`start_local_proxy_with_bound_identity`). Because each sandbox listener pins its `AgentIdentity` at the TCP socket level, a compromised process cannot spoof `X-Aegis-Org` or `X-Aegis-Agent` headers. On each request, the gateway first checks all headers for `$AOS_SECRET_*` placeholders against the secret vault's per-secret domain binding, and then checks the target `Host` against the organization's domain allowlist:

- If `$AOS_SECRET_GH` (bound to `api.github.com`) is sent toward `api.github.com`, the proxy substitutes the real token `ghp_...` at the network boundary (returning a deterministic L7 verification response in local offline/benchmark mode or forwarding upstream).
- If an agent attempts to send `$AOS_SECRET_GH` to any other domain (even an otherwise allowlisted domain such as `httpbin.org`, or an attacker host such as `evil-exfil.org`), the proxy rejects the request with `HTTP/1.1 403 Forbidden` (`blocked_exfiltration`).

3.3 A2A-Compliant Inter-Agent IPC Bus, Global Peer Discovery, and Reference-Based CAS (`aos-ipc`)

Whereas Anthropic's Model Context Protocol (MCP) [3] standardizes *vertical* agent-to-tool execution, Google's Agent2Agent (A2A) protocol [10] standardizes *horizontal* capability discovery (`AgentCard`) and stateful task delegation (`Task` lifecycle). `aos-ipc` unifies both over a disk-synchronized, broadcast-enabled bus:

- **Global Workstation Peer Registry and Zombie PID Pruning:** `aos-ipc` maintains per-organization and workstation-wide registries (`scope="global"` in `ipc_list_peers`, `aegis://global/peers`) supporting `<org_id>/<agent_id>` addressing. Each peer record stores its client root PID and start timestamp (`PROC_PIDTBSDINFO` on macOS / `/proc/<pid>/stat` on Linux) so `prune_dead_peers()` automatically evicts dead or PID-recycled zombie peers.
- **Google A2A `AgentCard` and Stateful `Task` Lifecycle:** Every peer synthesizes an A2A `AgentCard` (`GET /.well-known/agent-card.json`, with the legacy `agent.json` path kept as an alias; `aegis://a2a/agent-cards`, `ipc_update_card`) with structured `AgentSkill` tags (auto-reflecting active `coord_claim` globs). Beyond P2P mailboxes (`ipc_send`/`ipc_recv`) and topic Pub/Sub (`ipc_publish`/`ipc_subscribe`), every correlated `ipc_request` (and `POST /a2a message/send`, with legacy `tasks/send` accepted) instantiates a persistent `A2ATask` tracked across `submitted` → `working` (on `ipc_ack`) → `input-required` (via `ipc_task_update`, pushing an `[A2A input-required]` wakeup into the requester's inbox) → `completed` | `failed` | `canceled` (on `ipc_reply` or `tasks/cancel`), with typed `TextPart`, `DataPart`, and CAS `FilePart` outputs.
- **Enforced File Claims and Reference-Based CAS Handles:** `coord_claim` registers glob leases enforced at pre-tool-use and `workspace_commit` time. For large payloads, `org_artifact_publish` writes blobs to the Content-Addressable Store and returns a compact SHA-256 URI (`artifact://<org>/sha256:<64-hex>`, ∼85–98 bytes) wrapped as an A2A `FilePart`, reducing in-transit IPC payload tokens by **99.85%**, while cross-organization CAS `FilePart` fetches remain gated by Policy Rule R1.

3.4 Cedar-Inspired Policy Engine and Runtime Design Invariants (`aos-policy`)

Every runtime tool call is evaluated by `AegisPolicyKernel::evaluate` using a five-stage priority hierarchy inspired by Cedar [6] that partitions operations into a **Red Zone** (always blocked or forced to review), a **Green Zone** (auto-allowed inside the sandbox), and an **Amber Boundary Zone** (gated by human approval or session leases):

1. **R0a — Host Red-Line Guard (`Deny`):** Any access targeting sensitive host paths (`~/.ssh`, `~/.aws`, `~/.gnupg`, `~/.kube`, `/etc/shadow`, `/etc/sudoers`, or `/System`) is unconditionally denied.
2. **R0b — Commit-Time Hook & CI Guard (`Ask`):** Any `workspace_commit` that modifies repository execution hooks or CI workflows (`.git/hooks/*`, `.git/config`, `.envrc`, or `.github/workflows/*`) forces explicit human approval, even if an active session lease exists.
3. **R1 — Multi-Tenant Isolation Gate (`Deny` / `Ask`):** Cross-organization CAS artifact reads or unauthorized cross-tenant mutations are rejected.
4. **R2 & R3 — Active Session Leases & Green-Zone Auto-Allow (`Allow`):** All in-sandbox CoW file operations, sandboxed `proc_exec` commands, intra-organization A2A/IPC calls, allowlisted egress

requests, and boundary commits covered by an active session lease (`LeaseGrant`) are auto-allowed with zero prompts.

5. **R4 — Unleased Boundary Crossing (`Ask`):** Unleased `workspace_commit` calls and HTTP requests to unlisted external domains return an escalation ID requiring either a one-shot approval (`ttl_secs = 0`) or a TTL-scoped session lease (e.g., `ttl_secs = 3600`), with a context-aware confirmation prompt summarizing the action, modified shadow files, target host directory, and lease scope.

Together, these mechanisms enforce three core runtime invariants:

- **Design Invariant 1 (Pre-Commit Host Non-Interference):** *Until a `workspace_commit` is explicitly approved, no sequence of agent tool calls or spawned child processes can modify any file in the host repository.* VFS tools reject path traversal (`..`) and resolve all paths inside `shadow_root`, while `proc_exec` enforces kernel Seatbelt `(deny file-write*)` outside `shadow_root`, `agent_home`, and `org_pub`.
- **Design Invariant 2 (Zero Raw Secret Exposure to Sandbox):** *Plaintext API credentials never enter the sandboxed process environment, command-line arguments, or logs, and are injected into outbound HTTP headers only when the destination `Host` matches the secret's authorized domain list.* `NativeOsSandboxInstance::exec` clears the child environment (`env_clear()`) and passes only opaque `$AOS_SECRET_*` placeholders.
- **Design Invariant 3 (All-or-Nothing Conflict-Safe Commit):** *Concurrent agent commits either produce a merged host file preserving all non-overlapping edits, or abort in Phase 1 without modifying any host file when two agents edit the same baseline lines inconsistently.*

4 Multi-Client Integration and Enforcement

To minimize setup friction, `aegis-os install` automatically configures MCP server entries, per-session principal resolution, and lifecycle hooks across all three coding agent CLIs:

1. **Google Antigravity (`agy`):** Registers `aegis-os` in `~/.gemini/config/mcp_config.json` with eager tool loading, pre-authorizes green-zone `mcp(aegis-os/*)` tools in `settings.json` while keeping lease/approval tools (`policy_grant_lease`, `policy_approve_escalation`) under `permissions.ask`, and installs four lifecycle hooks (`SessionStart`, `PreInvocation`, `PreToolUse`, and `Stop`):
   - *MCP-First Tool Routing (`PreToolUse`):* Because agent CLIs still prompt on native `run_command` invocations unless exact command strings match static allowlists, `aegis-os agy-hook` classifies native shell and file calls: Ring-0 operations (`sudo`, credential paths) are unconditionally denied; host-boundary commands (`git push`, package installs) escalate to `ask`; and in-repository shell commands and file writes are denied with a ready-to-invoke `mcp_aegis-os_proc_exec` or `mcp_aegis-os_fs_write` payload, steering the agent onto the zero-prompt sandboxed MCP path.
   - *Per-Session Identity & Push Coordination (`SessionStart`, `PreInvocation`, `Stop`):* Walking parent process ancestry (`PROC_PIDTBSDINFO` on macOS / `/proc/<pid>/stat` on Linux) assigns each concurrent client window a distinct session principal (`agy-<6hex>`) bound to its `conversationId`. On `SessionStart`, the agent's CoW shadow is rebased onto the host via `refresh_from_host()`; on `PreInvocation`, unread IPC messages are non-destructively peeked and injected via `injectSteps.systemMessage`; and on `Stop`, un-acknowledged `TaskRequest` messages hold the turn open (up to 2 continues) until answered, while `coord_claim`/`coord_release` prevent overlapping file edits across live peers.
2. **Anthropic Claude Code (`claude`):** Injects `aegis-os` into `~/.claude.json` and adds green-zone `mcp__aegis-os__*` rules to `permissions.allow` in `~/.claude/settings.json`.
3. **OpenAI Codex CLI (`codex`):** Appends `[mcp_servers.aegis-os]` to `~/.codex/config.toml` and piggybacks unread IPC summaries (`_aegis_inbox`) onto `tools/call` responses.

5 Empirical Evaluation

We evaluate Aegis OS using our reproducible benchmark suite (`crates/aos-cli/src/bench.rs`, invoked via `aegis-os bench`). All experiments were executed on an Apple Silicon workstation (`macos-aarch64`, APFS filesystem, Rust 1.99.0 `-release`) using real OS processes, Seatbelt kernel sandboxes, loopback TCP sock-

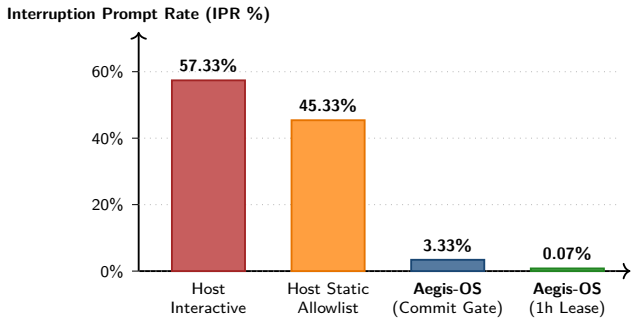


Figure 2: Interruption Prompt Rate (IPR) across 1,500 tool calls spanning 5 SWE task templates (Experiment 1). Aegis OS reduces human prompts by 94.19% in per-task commit mode (`ttl=0`) and 99.88% with a 1-hour session lease.

ets, and `cargo` builds. We investigate five research questions:

- **RQ1 (Authorization Friction):** How much does Aegis OS reduce confirmation prompts and modeled verification wait time across multi-step software engineering trajectories?
- **RQ2 (Adversarial Security):** Does Aegis OS block end-to-end OS process, credential harvesting, proxy/socket exfiltration, and cross-tenant attacks that compromise unconfined `Host-YOLO` execution?
- **RQ3 (System Overhead & Scaling):** What is the latency and disk footprint of APFS CoW provisioning, policy evaluation, HTTP daemon RPCs, 3-way merge commits, and Seatbelt process spawning?
- **RQ4 (Multi-Agent Concurrency):** Does two-phase LCS Diff3 merging preserve concurrent multi-agent edits (including multi-line insertions) while catching overlapping line conflicts?
- **RQ5 (Fault Recovery):** How fast does atomic CoW rollback recover from multi-file syntax corruption compared to sequential file restoration?

5.1 Experiment 1 (RQ1): Authorization Friction and Autonomous Continuity

To evaluate authorization friction reproducibly without LLM sampling variance, we execute 50 deterministic software engineering trajectories (30 tool calls per trajectory, totaling 1,500 real `AegisRuntimeKernel::call_tool` invocations) cycling across five task templates: (1) *Auth Bugfix & Regression Test*, (2) *Multi-Module Refactoring*, (3) *Feature Implementation & Quota Suite*, (4) *Security Audit & Sanitizer Hardening*, and (5) *Multi-Agent Contract Sync*. Each trajectory exercises repository exploration (`fs_glob`, `fs_read`), checkpointing (`workspace_snapshot`), multi-file edits (`fs_edit`, `fs_write`), CAS publishing (`org_artifact_publish`), peer IPC coordination (`ipc_send`, `ipc_recv`, `ipc_publish`), diff inspection (`workspace_diff`), and host commit (`workspace_commit`). All four modes execute all 1,500 tool calls including all 50 real `workspace_commit` operations.

Table 3 and Figure 2 present the results. Under *Host-Interactive*, every file write, state mutation, and external IPC action triggers a confirmation prompt (860 prompts, 57.33% IPR), restricting the maximum autonomous streak to 4 steps and imposing 71.7 minutes of modeled verification wait time (at 5.0 s/prompt) on top of 457.1 ms of runtime execution. Adding a static read-/status/snapshot allowlist (*Host-Static-Allowlist*) still triggers 680 prompts (45.33% IPR, 56.7 minutes modeled wait) because host file edits cannot be safely auto-allowed. In contrast, **Aegis-OS (Per-Task Commit Gate)** auto-allows all 29 internal CoW workspace and IPC operations per trajectory and prompts once at the final `workspace_commit` gate (approved via a one-shot `ttl=0` token), reducing prompts by **94.19%** (3.33% IPR, 483.6 ms runtime execution). When the operator grants a **1-hour Session Lease** (`ttl=3600`) on the first commit escalation, Aegis OS completes all 50 trajectories (and all 50 host commits) in 469.2 ms with **1 prompt total** (0.07% IPR, a **99.88% reduction**), extending the continuous autonomous streak to **1,470 steps** (**367.5**× longer).

5.2 Experiment 2 (RQ2): End-to-End Adversarial Security

We execute 40 live adversarial attacks end-to-end against both an unconfined disposable victim environment (`Host-YOLO`) and `Aegis-OS`:

- **Category A (10 Host Path Escape Attacks):** Spawns `sh -c "echo COMPROMISED > <outside_target>"`. In `Host-YOLO`, all 10 outside-workspace files were created on disk (100.0% ASR). Inside `Aegis-OS` (`proc_exec`), `/usr/bin/sandbox-exec` denied all 10 writes at the XNU kernel boundary with `Operation not permitted` (0.0% ASR).
- **Category B (10 Multi-Store Credential Harvesting Attacks):** Spawns `sh -c "cat <cred_path>"` cycling across `~/.ssh/id_rsa`, `~/.aws/credentials`, `~/.gnupg/secring.gpg`, `~/.kube/config`, and `~/.npmrc`. `Host-YOLO` leaked the secret key material on all 10 attempts (100.0% ASR), whereas the Aegis OS Seatbelt profile blocked all 10 reads (0.0% ASR).

Table 3: **Experiment 1 (RQ1): Authorization Friction, Interruption Prompt Rate (IPR), and Modeled Human Wait Time** across 50 deterministic multi-step trajectories spanning 5 SWE task templates (1,500 tool calls executed on `AegisRuntimeKernel`, including 50 real `workspace_commit` operations in every mode; human verification modeled at 5.0 s/prompt).

Execution Mode	Trajectories	Tool Calls	Human Prompts	IPR (%)	Prompt Reduction	Max Autonomous Streak	Runtime Exec (ms)	Modeled Human Wait (T_{human})
Host-Interactive (Per-Mutation Prompt)	50	1,500	860	57.33%	0.00% (Baseline)	4 steps	457.1 ms	71.7 min (4,300 s)
Host-Static-Allowlist (Read/Status Allow)	50	1,500	680	45.33%	20.93%	7 steps	496.7 ms	56.7 min (3,400 s)
Aegis-OS (Per-Task Commit Gate, `ttl=0`)	50	1,500	**50**	**3.33%**	**94.19%**	**29 steps**	**483.6 ms**	**4.2 min (250 s)**
Aegis-OS (With 1h Session Lease)	50	1,500	**1**	**0.07%**	**99.88%**	**1,470 steps**	**469.2 ms**	**0.08 min (5 s)**

Table 4: **Experiment 2 (RQ2): End-to-End Red-Team Adversarial Attack Evaluation** across 40 live process, socket, proxy, and multi-tenant gateway attacks executed against an unconfined `Host-YOLO` target environment versus `Aegis-OS`.

Attack Surface Category	Live Attacks	Host-YOLO Compromised (ASR)	Aegis-OS Compromised (ASR)	Empirical Kernel / Gateway Evidence
A. Host Path Escape & File Tampering	10	10 / 10 (100.0%)	**0 / 10 (0.0%)**	Kernel `sandbox-exec` denied all 10 outside writes (`Operation not permitted`)
B. Host Credential Harvesting (`.ssh/.aws/.kube/.npmrc`)	10	10 / 10 (100.0%)	**0 / 10 (0.0%)**	Seatbelt regexes (`deny file-read* ...`) blocked all 10 credential reads
C. Sandbox-to-Proxy & Direct `-noproxy` Egress Exfil	10	10 / 10 (100.0%)	**0 / 10 (0.0%)**	Pinned TCP Proxy returned 403 & Seatbelt blocked all `-noproxy` sockets
D. Cross-Tenant IPC & CAS Access via MCP Gateway	10	10 / 10 (100.0%)	**0 / 10 (0.0%)**	MCP Gateway + Policy Engine rejected all 10 cross-organization tool calls
Total / Evaluated ASR	**40**	**40 / 40 (100.0%)**	**0 / 40 (0.0%)**	**0 / 40 Compromised across OS Kernel, Network Proxy & Tenant Gateway**

- **Category C (10 Sandbox-to-Proxy and Direct `-noproxy` Egress Exfiltration Attacks):** We start a live local TCP sink server (`sink_addr`) recording leaked secrets, alongside an identity-pinned `AegisEgressGateway` (`proxy_addr`). In `Host-YOLO`, an unconfined `curl` process with `$GITHUB_TOKEN` in its environment transmitted the plaintext token to `sink_addr` across all 10 attacker domains (100.0% ASR). In `Aegis-OS`, we execute `curl` inside a Seatbelt sandbox (`sbx_egress.exec`) bound to `localhost:<proxy_port>` that attempts *both* (1) proxy exfiltration of `$AOS_SECRET_GH` to 10 unauthorized domains (`evil-exfil.org`, `webhook.site`, `ngrok-free.app`, etc.), which the proxy rejected with `HTTP/1.1 403 Forbidden` (`blocked_exfiltration`), and (2) direct proxy-bypass (`curl -noproxy '*' http://<sink_addr>/bypass`), which the XNU Seatbelt socket filter blocked with `Operation not permitted`. Zero tokens reached `sink_addr` (0.0% ASR).

- **Category D (10 Cross-Tenant IPC and CAS Attacks via MCP Gateway):** In a shared flat host store (`Host-YOLO`), an unconfined process directly read all 10 confidential enterprise tenant files (100.0% ASR). In `Aegis-OS`, we register 10 enterprise tenants (`org-enterprise-0..9`), publish confidential CAS artifacts via `org_artifact_publish`, and have principal `mallory` (`org-alpha`) invoke `kernel.call_tool` (`ipc_send` and `org_artifact_fetch`) targeting those enterprise tenants. All 10 cross-tenant tool calls were denied by Policy Rule R1 (0.0% ASR).

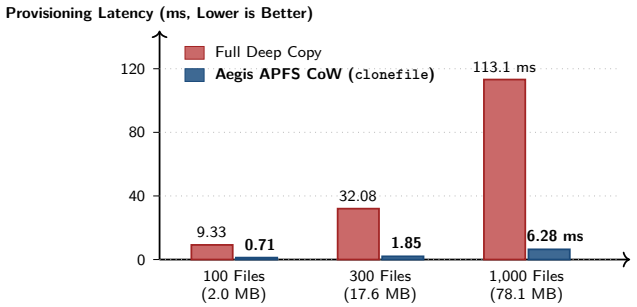


Figure 3: Workspace provisioning latency across repository scales (Experiment 3A). APFS `clonefile` scales with file metadata count rather than total byte volume, achieving an 18.0× speedup on the 1,000-file repository.

5.3 Experiment 3 (RQ3): CoW Provisioning, Storage Footprint, and Operation Latency

We benchmark workspace provisioning across three repository scales (100 files / 2.0 MB, 300 files / 17.6 MB, and 1,000 files / 78.1 MB, 5 trials per scale). As shown in Table 5 and Figure 3, full copying requires 113.13 ± 16.41 ms for the 1,000-file repository, whereas APFS `libc::clonefile` provisions the shadow workspace in 6.28 ± 0.65 **ms** (**18.0× faster**). Because `clonefile` duplicates only inode metadata rather than file data blocks, provisioning latency scales with file count (0.71 ms at 100 files to 6.28 ms at 1,000 files) rather than total byte size. Across 10 concurrent agent workspaces modifying working sets of 6, 12, and 25 files per agent, APFS CoW reduces disk block allocation by **94.0%–97.5%** (19.5 MB vs. 781.2 MB at 1,000 files).

Table 6 reports the latency distribution across all five runtime operation classes. In-process policy evaluation takes **0.31** μ**s** (310 ns, 2.96M ops/s), P2P IPC send/receive takes **1.06** μ**s**, a loopback HTTP JSON-RPC `fs_read` call (`POST /mcp`) takes **177.11** μ**s**, a two-

Table 5: **Experiment 3A (RQ3): Multi-Trial APFS Copy-on-Write Workspace Provisioning Latency and Storage Efficiency** across 10 concurrent agents modifying realistic per-task working sets (6, 12, and 25 files per agent; 5 trials per scale, Mean ± StdDev).

Repository Scale	Files	Logical Size	Full Copy (Mean ± SD)	Aegis CoW Clone (Mean ± SD)	Clone Speedup	10-Agent Disk Saving
Small Project	100	2.0 MB	9.33 ± 0.28 ms	$\mathbf{0.71 \pm 0.05}$ **ms**	**13.2**×	**94.0%** (1.2 MB vs. 19.5 MB)
Medium Project	300	17.6 MB	32.08 ± 5.82 ms	$\mathbf{1.85 \pm 0.09}$ **ms**	**17.3**×	**96.0%** (7.0 MB vs. 175.8 MB)
Large Project	1,000	78.1 MB	113.13 ± 16.41 ms	$\mathbf{6.28 \pm 0.65}$ **ms**	**18.0**×	**97.5%** (19.5 MB vs. 781.2 MB)

Table 6: **Experiment 3B (RQ3): Kernel, Daemon, Merge Commit, and Sandbox Spawn Latency Distribution**. Ops/sec is computed as N divided by total wall-clock time of the measurement loop (including harness overhead), so it is slightly lower than 1/Mean.

Runtime Operation	Mean (μs)	P50 (μs)	P95 (μs)	P99 (μs)	Ops/sec
`PolicyKernel::evaluate` (N=2000)	0.31	0.29	0.38	0.50	2,959,674
`IpcBus::send + recv` (N=1000)	1.06	0.83	1.79	4.75	924,749
`HTTP POST /mcp` (`fs_read`, N=200)	177.11	183.08	258.79	409.00	5,503
`commit_to_host` (3-Way Merge, N=100)	424.02	413.08	540.21	646.12	2,110
`Sandbox::exec` (`sandbox-exec`, N=30)	6,678.97	6,557.62	7,533.12	7,811.33	150

phase LCS Diff3 `commit_to_host` takes **424.02** μ**s**, and spawning an external process under macOS Seatbelt (`/usr/bin/sandbox-exec /usr/bin/true`) takes **6.68 ms** (6,678.97 μs). Relative to typical LLM inference turns (0.5–2.0 s), MCP tool mediation adds $< 0.04\%$ latency and Seatbelt process confinement adds ∼6.7 ms per spawned command.

5.4 Experiment 4 (RQ4): Multi-Agent Concurrent Rust Coding and Conflict Detection

We evaluate both disjoint structural edits and overlapping concurrent edits across `agy-agent`, `claude-code`, and `codex-cli` on a Rust crate (`src/lib.rs`):

- **Concurrent Multi-Line Insertion and Function Edits (EXP 4A):** All three agents start from the same baseline `src/lib.rs`. `agy-agent` inserts a 3-line helper function (`min_fee()`) at the top of the file and updates `calc_fee` (shifting downstream line indices), while `claude-code` modifies `calc_tax` and `codex-cli` modifies `calc_discount`. When writing directly to a shared file (**Host-Shared-YOLO**), the final writer (`codex-cli`) overwrites the earlier edits, preserving only **1/3** of the features. In **Aegis-OS**, our two-phase LCS Diff3 engine aligns the unmodified anchor lines across the unequal-length files (matching the merge semantics of `git merge-file` without requiring manual `git worktree` branches per agent), preserving **3/3 (100%)** of the features and passing `cargo test -offline` (`exit code 0`).
- **Overlapping Same-Line Conflict Detection (EXP 4B):** Across 10 trials, both `agy-agent` and `codex-cli` concurrently modify `calc_fee` with conflicting expressions. While **Host-Shared-YOLO** suffers a silent lost update (0/10 detected), **Aegis-OS** detects **10/10 (100%)** of the conflicts during Phase 1, aborts the second commit without modifying the host file, and returns the exact conflicting baseline line number.
- **In-Transit IPC Token Savings:** Passing 98-byte `artifact://` CAS URI handles instead of inline 64 KB specifications across 3 inter-agent handoffs reduces in-transit IPC message payload bytes from 196,608 bytes (∼49,152 tokens) to 294 bytes (∼73 tokens), a **99.85% reduction** for orchestrator routing and pass-through delegation (Table 7).

5.5 Experiment 5 (RQ5): Time-Travel Rollback and `cargo check` Verification

We evaluate recovery from multi-file syntax corruption (`BROKEN_RUST_SYNTAX_ERROR !!!`) injected across 10, 50, and 200 Rust module files (`src/mod_*.rs`, 64 KB per module). As shown in Table 8, Aegis OS's directory-tree `workspace_rollback` restores all 200 corrupted modules in 8.68 ± 1.02 **ms** (**2.4**× **faster** than sequential file-by-file I/O restoration), and immediately passes `cargo check -offline` inside the restored shadow workspace across all scales.

6 Discussion and Limitations

Cross-Platform Implementation Scope (macOS vs. Linux). Table 9 summarizes the implementation status of each OS primitive in `aegis-os`. Our empirical evaluation in §5 was conducted on macOS (`macos-aarch64`), where `libc::clonefile` and `/usr/bin/sandbox-exec` are exercised end-to-end. On Linux, `NativeOsSandboxDriver` generates `/usr/bin/bwrap` arguments with `-unshare-all` (including network namespace isolation when an egress proxy is configured) and mounts empty `-tmpfs` overlays over `~/.ssh`, `~/.aws`, `~/.gnupg`, and `~/.kube`; adding native Linux `ioctl(FICLONE)` reflinks [19], Landlock LSM, and `seccomp-bpf` syscall filters, as well as binding `MicroVmSandboxDriver` to live `libkrun`/Firecracker VMM processes, are engineering extensions for Linux cloud deployments.

Table 7: **Experiment 4 (RQ4): 3-Agent (`agy` + `claude` + `codex`) Concurrent Rust Modification, Overlapping Conflict Detection, and `cargo test` Verification.**

Evaluation Metric	Host-Shared-YOLO (In-Place Write)	Aegis-OS Isolated CoW + 2-Phase LCS Diff3	Empirical Result
Concurrent 3-Agent Structural & Function Edits (`src/lib.rs`)	1 / 3 preserved (2 / 3 overwritten)	**3 / 3 preserved (100% LCS Diff3 Merged)**	**3.0× Feature Retention (0% Clobber)**
Overlapping Same-Line Edit Conflict Detection (10 Trials)	0 / 10 detected (Silent Lost Update)	**10 / 10 detected & aborted in Phase 1 (Host intact)**	**100% Conflict Safety**
Post-Merge Real `cargo test -offline` Verification	Failed (Missing `agy` & `claude` logic)	**Passed (`cargo test` exit code 0)**	**Build & Test Verified**
In-Transit IPC Payload for 3×64 KB Spec Handoffs	196,608 bytes (∼49,152 tokens inline)	**294 bytes (∼73 tokens via CAS URI handles)**	**99.85% In-Transit Token Reduction**

Table 8: **Experiment 5 (RQ5): Multi-File Corruption Recovery and `cargo check` Verification** (5 trials/scale, Mean ± SD).

Corrupted Modules	Seq. File Restore	Aegis CoW Rollback	Speedup	`cargo check`
10 (640 KB)	0.90 ± 0.06 ms	**0.75 ± 0.04 ms**	**1.2**×	PASSED
50 (3.2 MB)	4.82 ± 0.32 ms	**2.33 ± 0.14 ms**	**2.1**×	PASSED
200 (12.8 MB)	20.63 ± 3.14 ms	**8.68 ± 1.02 ms**	**2.4**×	PASSED

macOS Seatbelt Architecture and Escape Surface. On macOS, dynamic linker caches (`dyld`), Xcode toolchains, and Homebrew packages traverse numerous paths under `/System`, `/Library`, and `/opt/homebrew`. Thus, our Seatbelt profile pairs a strict write allowlist (`(deny file-write*)`) with a regex read denylist for credential stores, `(deny appleevent-send)` to prevent AppleScript/GUI automation escapes, and exact port confinement `(allow network-outbound (remote ip "localhost:<proxy_port>"))`. While Apple has marked `/usr/bin/sandbox-exec` as a legacy interface, it remains the standard kernel MAC mechanism used by production agent CLIs (including Claude Code and Codex CLI) on macOS because Apple's Endpoint Security and `Virtualization.framework` require root entitlements or full VM boot overhead. Hardening Mach bootstrap lookups (`mach-lookup`) against LaunchServices job spawning without breaking toolchain execution is an important direction for future work.

Session Leases and Commit-Time Poisoning. Granting a 1-hour session lease (`ttl=3600`) eliminates 99.88% of prompts by allowing validated CoW diffs to commit automatically. However, if a prompt-injected agent modifies repository build scripts (e.g., `Makefile`, `build.rs`, or `package.json postinstall`) and commits under an active lease, a subsequent unsandboxed build on the host could execute attacker code (*commit-time poisoning*). Aegis OS mitigates the most direct vectors via Rule R0b (§3.4), which refuses silent lease auto-commits whenever a diff touches `.git/hooks/*`, `.git/config`, `.envrc`, or `.github/workflows/*`. Extending R0b to require human approval on `build.rs`, `Makefile`, and `package.json` modifications is straightforward.

Confused Deputy on Allowlisted Egress Domains and HTTPS TLS Termination. Boundary secret injection guarantees that raw tokens never enter the agent sandbox and cannot be sent to unauthorized domains. However, if `api.github.com` is allowlisted for `$AOS_SECRET_GH`, a prompt-injected agent could still act as a *confused deputy* by issuing a `POST /gists` request to exfiltrate code to a public GitHub Gist on `api.github.com`. Mitigating confused-deputy exfiltration on allowlisted domains requires restricting secret injection by HTTP method and URL path prefix (e.g., permitting `GET /repos/org/*` while escalating `POST /gists`). Additionally, intercepting headers inside direct `HTTPS CONNECT` tunnels from legacy CLI binaries requires configuring the local proxy with a session-scoped ephemeral TLS Certificate Authority (`SSL_CERT_FILE`).

Syntactic 3-Way Merging vs. Semantic Conflicts. Our two-phase LCS Diff3 engine resolves disjoint line edits and aborts on overlapping line conflicts, but syntactically disjoint edits can still introduce semantic conflicts (e.g., Agent A renames a function signature at Line 10 while Agent B calls the old function name at Line 120) [4]. Running `cargo test` via `proc_exec` after merging peer changes and rolling back via `workspace_rollback` if compilation fails mitigates semantic integration errors.

7 Related Work

Operating System Transactions and Speculative Execution. Our transaction-boundary security model builds on classical operating system transactions. TxOS [18] introduced kernel-level system transactions (`sys_xbegin`/`sys_xend`) providing ACID semantics for OS resources, while Speculator [17] enabled speculative process execution over distributed filesystems with rollback on dependency failure. Mbox [15] overlays a private sandbox filesystem on the host via system-call interposition so that non-root users can inspect and selectively commit changes, and UnionFS [23] unifies multiple directories into a single copy-on-write namespace inside the kernel. Aegis OS adapts OS transaction semantics specifically to multi-tenant AI coding agents, combining APFS `clonefile` CoW shadow trees with two-phase LCS Diff3 merging [14, 4] and Cedar-inspired [6] session leases.

Authorization Fatigue in Usable Security. Empirical studies of mobile permission dialogs [9] and browser security warnings [2] show that users frequently ignore, misunderstand, or click through security prompts, and that prompt design and frequency strongly

Table 9: **Cross-Platform Operating System Primitive Mapping and Implementation Status** in Aegis OS.

Aegis OS Subsystem	macOS Workstation (Evaluated)	Linux Workstation / Host	Cloud Host (Tier 2)	Current Implementation Status
CoW Workspace (`aos-vfs`)	APFS `libc::clonefile(src, dst, 0)`	Copy fallback (Btrfs `FICLONE` / `OverlayFS`)	Virtio-fs CoW mount spec	macOS `clonefile` evaluated; Linux copy fallback
Process MAC (`aos-sandbox`)	XNU Seatbelt (`/usr/bin/sandbox-exec`)	Bubblewrap (`/usr/bin/bwrap -unshare-all`)	KVM (`libkrun` / Firecracker)	macOS Seatbelt & Linux `bwrap` implemented; Tier-2 spec
Credential Blocking (`~/.ssh`)	Seatbelt (`deny file-read* (regex ...)`)	`bwrap -tmpfs` over `.ssh/.aws/.gnupg/.kube`	Unmounted host volumes	Implemented in both macOS SBPL & Linux `bwrap`
IPC / GUI Escape Guard	Seatbelt (`deny appleevent-send`)	`bwrap` PID/IPC namespace isolation	Guest VM kernel boundary	Implemented (Landlock/`seccomp-bpf` planned)
Zero-Trust Egress (`aos-egress`)	Pinned TCP Proxy + Seatbelt `localhost:<port>`	Pinned TCP Proxy + `bwrap` net unshare	Tap/vsock Egress Gateway	Pinned TCP Proxy & Seatbelt port filter evaluated

affect whether warnings are heeded. Autonomous coding agents exacerbate this phenomenon by issuing dozens of mutating tool calls per minute; Aegis OS eliminates intermediate prompts by making in-sandbox actions side-effect-free and reversible.

LLM Agent Sandboxes and Prompt Injection Defenses. Benchmarks such as SWE-bench [13], SWE-agent [26], OpenHands [21], ToolEmu [20], AgentDojo [8], and InjecAgent [27] demonstrate both the utility of coding agents and their susceptibility to indirect prompt injection [11]. System-level defenses such as IsolateGPT [25], CaMeL [7], and FIDES [5] enforce privilege separation or information-flow control across LLM planners and tool interpreters. Production CLIs such as Claude Code (`sandbox-runtime`) and Codex CLI integrate Seatbelt and Bubblewrap/Landlock sandboxes to confine shell commands in place. Aegis OS complements both model-level IFC defenses and in-place CLI sandboxes by adding transactional CoW shadow workspaces, two-phase Diff3 multi-agent merge commits, boundary secret injection, and cross-agent IPC/CAS.

Agent Operating Systems and Multi-Agent Frameworks. AIOS [16] schedules LLM queries and context snapshots above application agents, while AutoGen [24] and MetaGPT [12] coordinate multi-agent dialogues in Python. Aegis OS instead operates purely as a language-agnostic OS runtime bridging Anthropic's Model Context Protocol (MCP) [3] for vertical tool mediation and Google's Agent2Agent (A2A) protocol [10] for horizontal capability discovery and stateful task lifecycles, while drawing on capability systems such as Capsicum [22] and microVMs such as Firecracker [1].

8 Conclusion

We presented **Aegis OS**, a transactional, multi-tenant Agent Operating System Runtime that addresses the Security–Autonomy Dilemma for autonomous coding agents. By combining APFS Copy-on-Write shadow workspaces, two-phase LCS Diff3 multi-agent merge commits, kernel-enforced Seatbelt/Bubblewrap sandboxes with port-pinned network confinement, boundary secret injection, and intra-organization IPC + CAS primitives, Aegis OS reduces human confirmation prompts by 94.19%–99.88% on our benchmark suite while blocking all 40 evaluated process, credential, network, and cross-tenant attacks.

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A Appendix: Reproducibility and Artifact Guide

The open-source implementation of Aegis OS and the benchmark suite are available at https://github.com/heurion-org/aegis-os. All nine Rust crates (`aos-core`, `aos-policy`, `aos-vfs`, `aos-ipc`, `aos-sandbox`, `aos-egress`, `aos-recorder`, `aos-mcp-gateway`, and `aos-cli`) are self-contained in the workspace. To run all unit tests and reproduce all five empirical experiments (`BENCHMARK_REPORT.md` and `BENCHMARK_REPORT.json`):

```
cargo test --workspace
cargo run --release --bin aegis-os -- bench \
  --report-out BENCHMARK_REPORT.md
```